DATA CATALOG

GOLD LAYER · DATA WAREHOUSE DOCUMENTATION

Overview

The **Gold Layer** represents the business-oriented layer of the Data Warehouse. It contains clean, structured, and analytics-ready data designed to support business intelligence (BI), reporting, and analytical use cases.

The Gold Layer follows a dimensional data modeling approach and consists of **dimension tables**, which store descriptive information about business entities, and **fact tables**, which store measurable business events and transactional data.

1. gold.dim_customers

Purpose: Stores customer information enriched with demographic and geographic attributes. This dimension enables customer segmentation and analysis based on customer characteristics.

Column Name	Data Type	Description
customer_key	INT	Surrogate key that uniquely identifies each customer record in the dimension table.
customer_id	INT	Unique numerical identifier assigned to the customer.
customer_number	NVARCHAR(50)	Alphanumeric customer identifier used for tracking and referencing.
first_name	NVARCHAR(50)	Customer's first name.
last_name	NVARCHAR(50)	Customer's last name or family name.
country	NVARCHAR(50)	Country of residence of the customer (e.g., Australia).
marital_status	NVARCHAR(50)	Customer's marital status (e.g., Married, Single).
gender	NVARCHAR(50)	Customer's gender (e.g., Male, Female, n/a).
birthdate	DATE	Customer's date of birth (YYYY-MM-DD).
create_date	DATE	Date when the customer record was created in the source system.

2. gold.dim_products

Purpose: Stores product-related information and descriptive attributes. This dimension enables product analysis based on categories, subcategories, product lines, and other characteristics.

Column Name	Data Type	Description
product_key	INT	Surrogate key that uniquely identifies each product record in the dimension table.
product_id	INT	Unique identifier assigned to the product for internal tracking.
product_number	NVARCHAR(50)	Structured alphanumeric product code used for identification and referencing.
product_name	NVARCHAR(50)	Descriptive product name, including details such as type, color, and size.
category_id	NVARCHAR(50)	Identifier linking the product to its corresponding category.
category	NVARCHAR(50)	High-level product classification (e.g., Bikes, Components).
subcategory	NVARCHAR(50)	More detailed classification of the product within its category.
maintenance	NVARCHAR(50)	Indicates whether the product requires maintenance (e.g., Yes, No).

Column Name	Data Type	Description
cost	INT	Product cost or base price, expressed in monetary units.
product_line	NVARCHAR(50)	Product line or series to which the product belongs (e.g., Road, Mountain).
start_date	DATE	Date when the product became available for sale or use.

3. gold.fact_sales

Purpose: Stores transactional sales data at the sales line-item level. It connects customer and product dimensions with measurable sales information, enabling business performance analysis and reporting.

Column Name	Data Type	Description
order_number	NVARCHAR(50)	Unique alphanumeric identifier for each sales order (e.g., SO54496).
product_key	INT	Foreign key linking the sales record to the product dimension.
customer_key	INT	Foreign key linking the sales record to the customer dimension.
order_date	DATE	Date when the customer placed the order.
shipping_date	DATE	Date when the order was shipped to the customer.
due_date	DATE	Date when payment for the order was due.
sales_amount	INT	Total monetary value of the sales line item, expressed in whole currency units.
quantity	INT	Number of product units ordered for the sales line item.
price	INT	Price per unit of the product, expressed in whole currency units.

Data Model Overview

Table	Table Type	Description
gold.dim_customers	Dimension	Customer attributes and demographic information.
gold.dim_products	Dimension	Product attributes and classification information.
gold.fact_sales	Fact	Transactional sales data and measurable business metrics.

Relationships

The `gold.fact_sales` table connects to both dimension tables through surrogate keys:

- fact_sales.customer_key → dim_customers.customer_key
- fact_sales.product_key → dim_products.product_key

This structure follows a **Star Schema** approach, making the data model suitable for analytical queries, aggregations, and BI reporting.

Intended Use Cases

- Customer segmentation and demographic analysis
- Product performance and category analysis
- Sales performance monitoring
- Revenue and quantity analysis
- Order and shipping analysis
- Customer purchasing behavior

Documentation note: This catalog describes the documented Gold Layer tables and columns. Data types and definitions should be validated against the actual database schema.